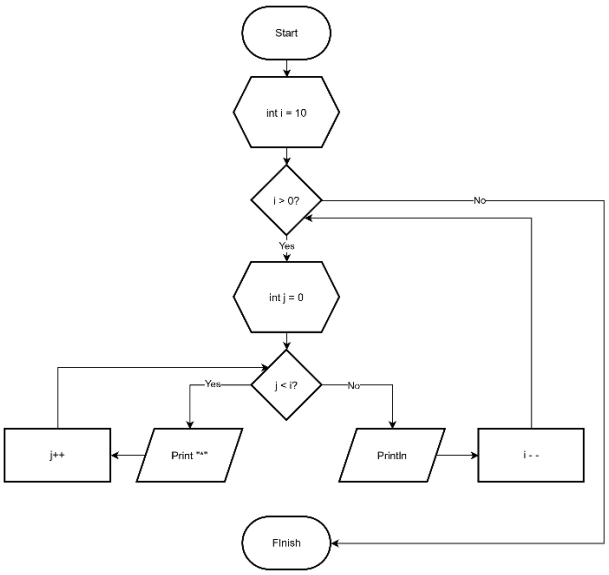
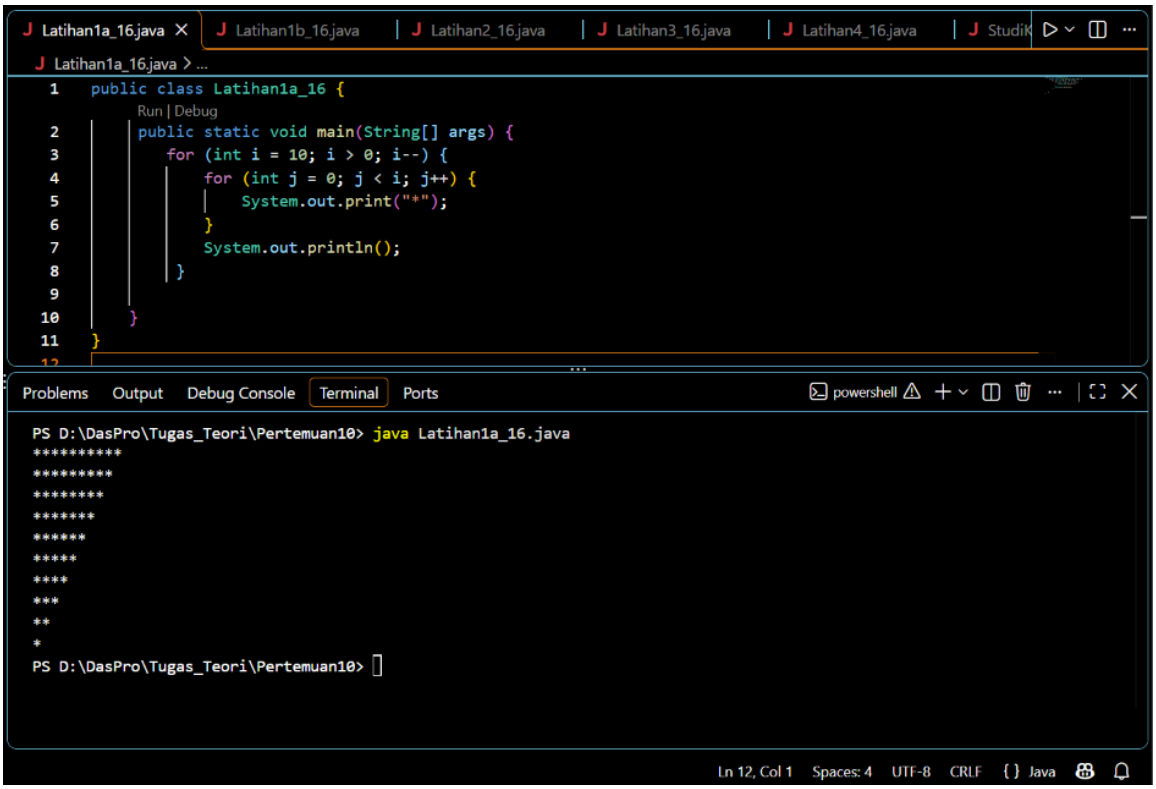
Dasar Pemrograman – Perulangan 2 (Latihan Soal)

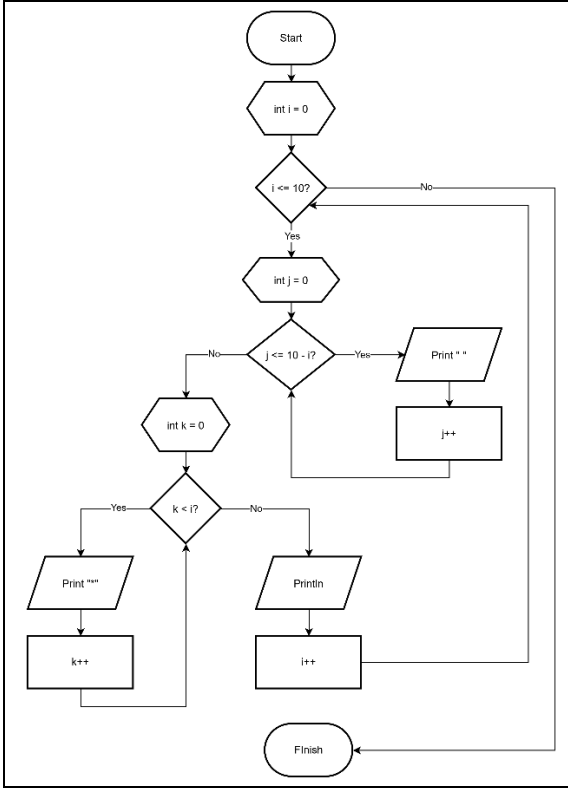
1. Latihan1a

Flowchart	Pseudocode
	**PROGRAM** Latihan1a_16 **DEKLARASI** i, j: int **ALGORITMA** FOR i = 10; i > 0; i - - DO FOR j = 0; j < i; j++ DO OUTPUT "*" tanpa pindah baris END FOR OUTPUT pindah baris END FOR END

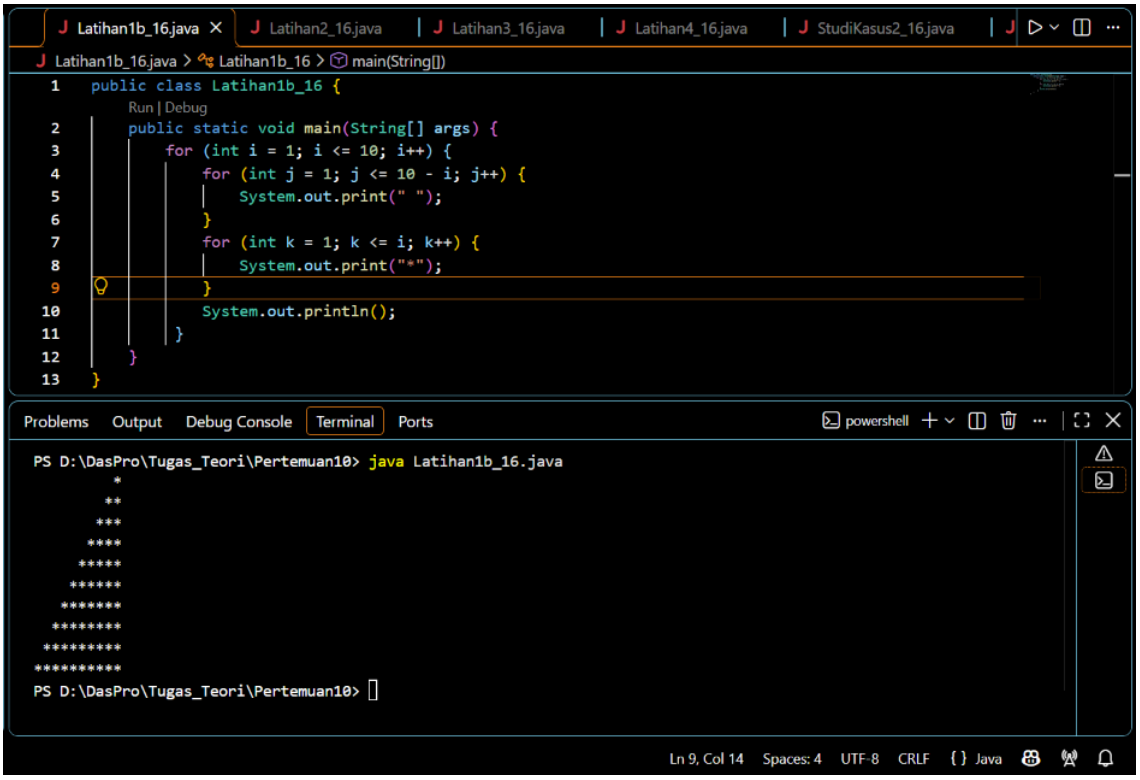
Program



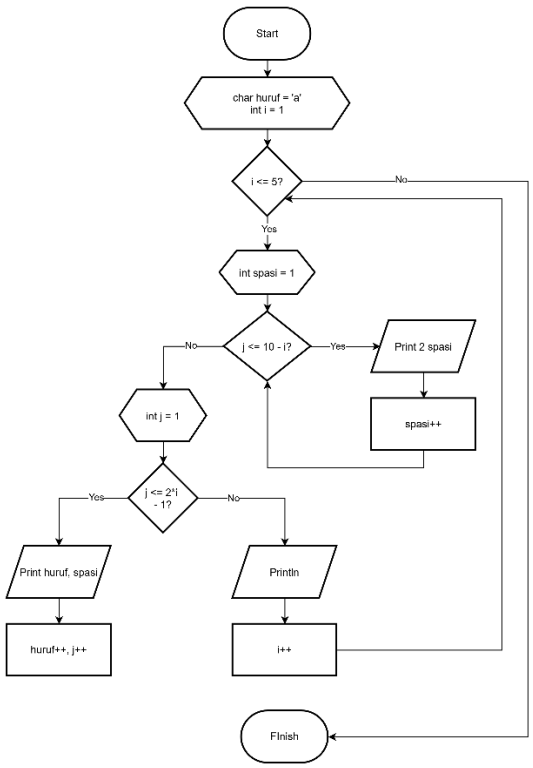
Latihan1b

Flowchart	Pseudocode
	**PROGRAM** Latihan1b_16 **DEKLARASI** i, j, k: int **ALGORITMA** FOR i = 0; i < 10; i++ DO FOR j = 0; j <= 10 - i; j++ DO OUTPUT " " tanpa pindah baris END FOR FOR k = 0; k < i; k++ DO OUTPUT "*" tanpa pindah baris END FOR OUTPUT pindah baris END FOR END

Program



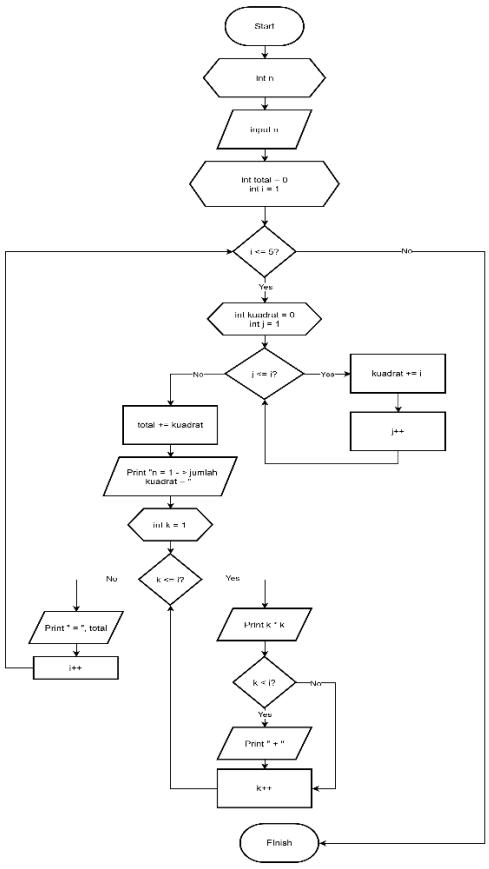
2.

Flowchart	Pseudocode
	**PROGRAM** Latihan2_16 **DEKLARASI** huruf = ’a’: char i, spasi, j: int **ALGORITMA** FOR i = 1; i <= 5; i++ DO FOR spasi = 1; spasi <= 5 - i; spasi++ DO OUTPUT " " tanpa pindah baris END FOR FOR j = 1; j <= 2*i - 1; j++ DO OUTPUT huruf, " " tanpa pindah baris huruf++ END FOR OUTPUT pindah baris END FOR END

Program

```
Latihan1b_16.java | Latihan2_16.java | Latihan3_16.java | Latihan4_16.java | StudiKasus2_16.java
Latihan2_16.java > ...
1   public class Latihan2_16 {
        Run | Debug
2       public static void main(String[] args) {
3       char huruf = 'a';
4           int baris = 5;
5
6           for (int i = 1; i <= baris; i++) {
7               for (int s = 1; s <= baris - i; s++) {
8                   System.out.print(" ");
9               }
10              int jumlahHuruf = 2 * i - 1;
11              for (int j = 1; j <= jumlahHuruf; j++) {
12                  System.out.print(huruf);
13                  huruf++;
14              }
15              System.out.println();
16          }
17      }
18  }
Problems  Output  Debug Console  Terminal  Ports        powershell
PS D:\DasPro\Tugas_Teori\Pertemuan10> java Latihan2_16.java
    a
   bcd
  efghi
 jklmnop
qrstuvwxy
PS D:\DasPro\Tugas_Teori\Pertemuan10>
```

3.

Flowchart	Pseudocode
	**PROGRAM** Latihan3_16 **DEKLARASI** n, total, i, j, kuadrat, k: int **ALGORITMA** INPUT n total = 0 FOR i = 1; i <= n; i++ DO kuadrat = 0 FOR j = 1; j <= i; j++ DO kuadrat = kuadrat + i END FOR total += kuadrat OUTPUT "n = ", i, " -> jumlah kuadrat = " FOR k = 1; k <= i; k++ DO OUTPUT k*k tanpa pindah baris IF k < i THEN OUTPUT " + " tanpa pindah baris END FOR OUTPUT " = ", total END FOR END

Program

```java
import java.util.Scanner;
public class Latihan3_16 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Masukkan nilai n: ");
        int n = input.nextInt();

        int total = 0;

        for (int i = 1; i <= n; i++) {
            int kuadrat = 0;

            for (int j = 1; j <= i; j++) {
                kuadrat += i;
            }

            total += kuadrat;

            System.out.print("n = " + i + " -> jumlah kuadrat = ");

            for (int k = 1; k <= i; k++) {
                System.out.print(k * k);

                if (k < i) {
                    System.out.print(" + ");
                }
            }

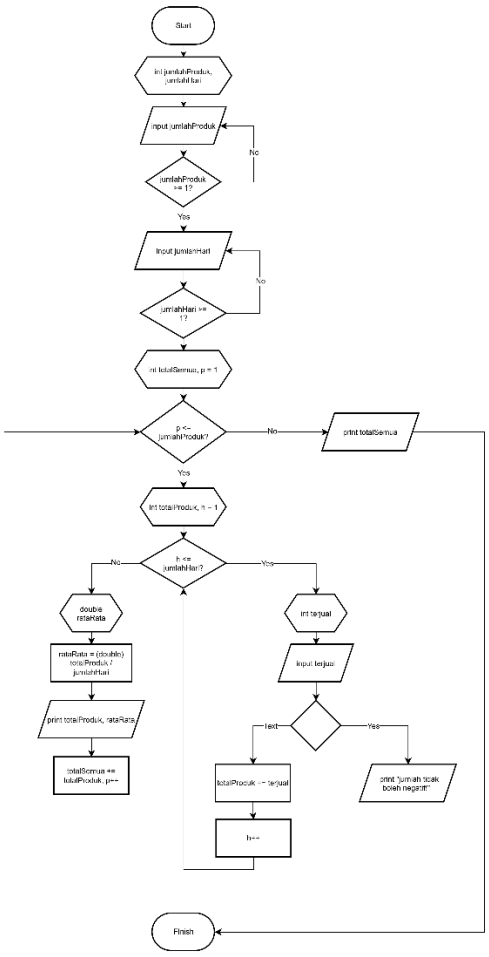
            System.out.println(" = " + total);

        input.close();
        }
    }
}
```

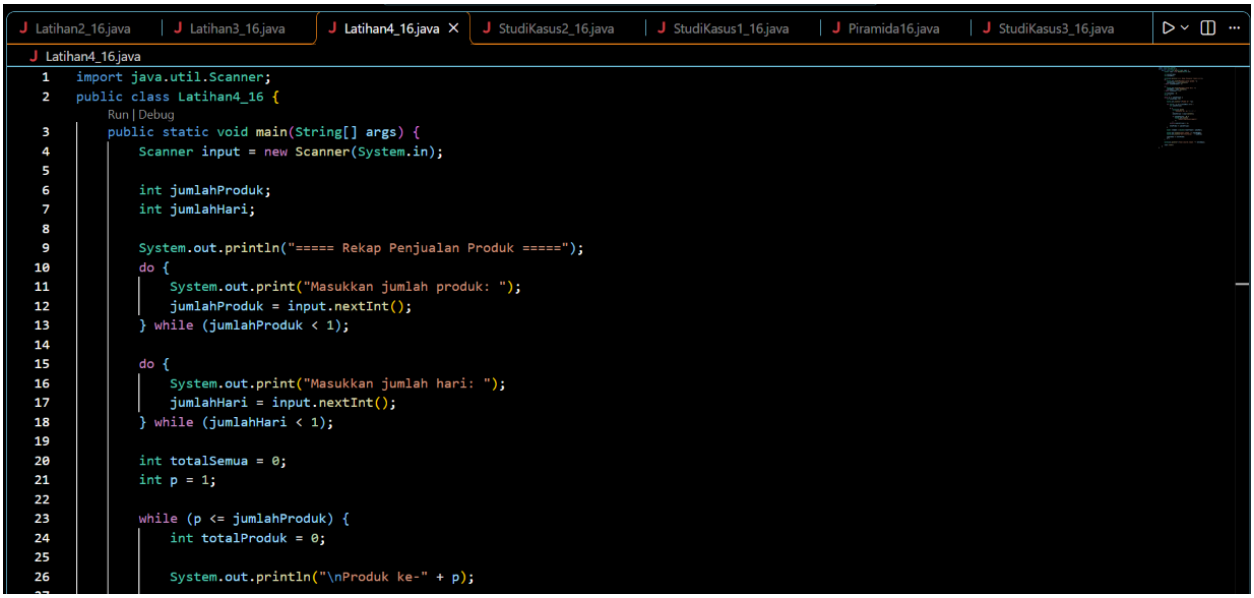
```
Problems  Output  Debug Console  Terminal  Ports

PS D:\DasPro\Tugas_Teori\Pertemuan10> java Latihan3_16.java
Masukkan nilai n: 5
n = 1 -> jumlah kuadrat = 1 = 1
n = 2 -> jumlah kuadrat = 1 + 4 = 5
n = 3 -> jumlah kuadrat = 1 + 4 + 9 = 14
n = 4 -> jumlah kuadrat = 1 + 4 + 9 + 16 = 30
n = 5 -> jumlah kuadrat = 1 + 4 + 9 + 16 + 25 = 55
PS D:\DasPro\Tugas_Teori\Pertemuan10>
```

4.

Flowchart	Pseudocode
	**PROGRAM** Latihan4_16 **DEKLARASI** jumlahProduk, jumlahHari, p, h, terjual, totalProduk, totalSemua: int rataRata: double **ALGORITMA** INPUT jumlahProduk WHILE jumlahProduk < 1 DO INPUT jumlahProduk END WHILE INPUT jumlahHari WHILE jumlahHari < 1 DO INPUT jumlahHari END WHILE p = 1 WHILE p <= jumlahProduk DO PRINT "Produk " + p totalProduk = 0 h = 1 WHILE h <= jumlahHari DO INPUT terjual WHILE terjual < 0 DO PRINT "Jumlah tidak boleh negatif!" INPUT terjual END WHILE totalProduk = totalProduk + terjual h = h + 1 END WHILE rataRata = totalProduk / jumlahHari PRINT "Total produk " + p + " = " + totalProduk PRINT "Rata-rata produk " + p + " = " + rataRata totalSemua = totalSemua + totalProduk p = p + 1 END WHILE PRINT "Total seluruh produk = " + totalSemua END

Program



```java
import java.util.Scanner;
public class Latihan4_16 {
    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int jumlahProduk;
        int jumlahHari;

        System.out.println("===== Rekap Penjualan Produk =====");
        do {
            System.out.print("Masukkan jumlah produk: ");
            jumlahProduk = input.nextInt();
        } while (jumlahProduk < 1);

        do {
            System.out.print("Masukkan jumlah hari: ");
            jumlahHari = input.nextInt();
        } while (jumlahHari < 1);

        int totalSemua = 0;
        int p = 1;

        while (p <= jumlahProduk) {
            int totalProduk = 0;

            System.out.println("\nProduk ke-" + p);
```



```java
        for (int h = 1; h <= jumlahHari; h++) {
            int jumlahTerjual;

            do {
                System.out.print(
                    "Penjualan hari ke-" + h + ": "
                );
                jumlahTerjual = input.nextInt();

                if (jumlahTerjual < 0) {
                    System.out.println(
                        "Jumlah tidak boleh negatif!"
                    );
                }
            } while (jumlahTerjual < 0);

            totalProduk += jumlahTerjual;
        }

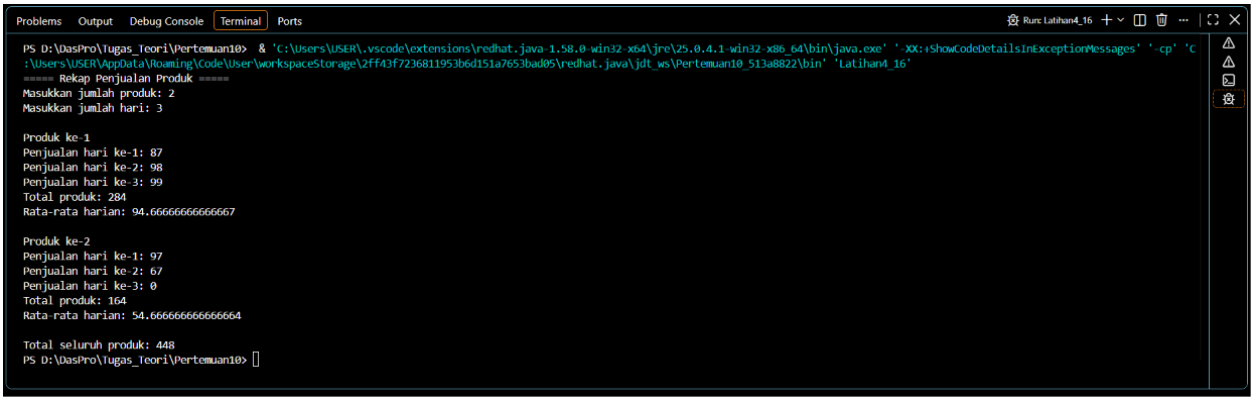
        double rataRata = (double) totalProduk / jumlahHari;

        System.out.println("Total produk: " + totalProduk);
        System.out.println("Rata-rata harian: " + rataRata);

        totalSemua += totalProduk;
        p++;
    }

    System.out.println("\nTotal seluruh produk: " + totalSemua);

    input.close();
```



```
PS D:\DasPro\Tugas_Teori\Pertemuan10> & 'C:\Users\USER\.vscode\extensions\redhat.java-1.58.0-win32-x64\jre\25.0.4.1-win32-x86_64\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\USER\AppData\Roaming\Code\User\workspaceStorage\2ff43f7236811953b6d151a7653bad05\redhat.java\jdt_ws\Pertemuan10_513a8822\bin' 'Latihan4_16'
===== Rekap Penjualan Produk =====
Masukkan jumlah produk: 2
Masukkan jumlah hari: 3

Produk ke-1
Penjualan hari ke-1: 87
Penjualan hari ke-2: 98
Penjualan hari ke-3: 99
Total produk: 284
Rata-rata harian: 94.66666666666667

Produk ke-2
Penjualan hari ke-1: 97
Penjualan hari ke-2: 67
Penjualan hari ke-3: 0
Total produk: 164
Rata-rata harian: 54.666666666666664

Total seluruh produk: 448
PS D:\DasPro\Tugas_Teori\Pertemuan10>
```